

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 2: Communication

Subtopic 2.1: Networks including the Internet

Concept 2.1.2: Network Models (Client-Server, Peer-to-Peer)

Total Questions: 4

Mark Schemes begin on Page 4

Question 1 | Source: 9618_s25_qp_13 (Q3.c)

3 A programmer is writing a program in a high-level language.

(c) The computer program allows users to play a game in a virtual world. Users can play the game from any computer with internet access using a web browser. The game allows users to interact with other users.

Explain the reasons why the statement ‘This computer game uses a client-server model’ is correct.

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Question 2 | Source: 9618_s24_qp_11 (Q5.a)

5 A bank allows customers to access their accounts using an application that they can download onto a device such as a smartphone.

(a) The system that allows customers to access their accounts using the application is a client-server model.

Describe the roles of the different devices in this model.

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Question 3 | Source: 9618_s21_qp_11 (Q4.a)

4 Melinda and her friends set up a peer-to-peer network between their computers to share data.

(a) Describe the key features of a peer-to-peer network.

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Question 4 | Source: 9618_s21_qp_11 (Q4.b)

4 Melinda and her friends set up a peer-to-peer network between their computers to share data.

(b) Describe **two** drawbacks to Melinda and her friends of using a peer-to-peer network.

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MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_s25_ms_13 (Q3.c)

3(c)	1 mark each to max 4 • A webserver stores all the data for each player • Each player is on a client computer // The player's web browser is the client • that sends requests over the internet to the web server • the server performs the required action in the game • the server updates the data in the game • the server sends the results to the player	4
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Mark Scheme for Question 2 | Source: 9618_s24_ms_11 (Q5.a)

5(a)	1 mark each: • Identification of server in the bank scenario • Description e.g. Receives requests, processes the requests • Identification of client in bank scenario • Description e.g. Sends request to the server, waits and outputs the response	4
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Mark Scheme for Question 3 | Source: 9618_s21_ms_11 (Q4.a)

4(a)	1 mark per bullet point to max 2 • All computers are of equal status • Each computer provides access to resources and data // data is distributed • Computers can communicate and share resources • Each computer is responsible for its own security	2
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Mark Scheme for Question 4 | Source: 9618_s21_ms_11 (Q4.b)

4(b)	1 mark per bullet point to max 2 per drawback • Reduced security // no central management of security • ... only as secure as the weakest computer on the network • ... each computer is at risk from viruses from other computers • No central management of backup • ... if the data from one computer is not backed up it is lost to all of them • No central management of files/software • ... consistency may be difficult to maintain • ... each computer may have different software from the others • Individual computers may respond slower • ... because they are being accessed by other computers • In order to share files etc. all the computers involved need to be switched on • ... so the files etc. may not be always available	4
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